

Taylor and Maclaurin Series

ANSWER KEY



The Taylor series formula

- 1 State the general Taylor series formula for $f(x)$ centered at $x = a$.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots$$

- 2 Find the Taylor series for $f(x) = \ln x$ centered at $a = 1$, up through the $(x-1)^3$ term.

$$f(1) = 0; f'(x) = \frac{1}{x}, f'(1) = 1; f''(x) = -\frac{1}{x^2}, f''(1) = -1; f'''(x) = \frac{2}{x^3}, f'''(1) = 2. \text{ Series: } 0 + (x-1) - \frac{1}{2}(x-1)^2 + \frac{2}{6}(x-1)^3 = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}.$$

Common Maclaurin series

- 3 State the Maclaurin series for e^x , $\sin x$, and $\cos x$.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

- 4 Use the Maclaurin series for e^x to find the Maclaurin series for e^{-x^2} .

$$\text{Substitute } x \rightarrow -x^2: e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} = 1 - x^2 + \frac{x^4}{2} - \dots$$

Using known series to build new series

- 5 Use the Maclaurin series for $\sin x$ to find the first three nonzero terms of the Maclaurin series for $x \sin x$.

$$x \sin x = x \left(x - \frac{x^3}{6} + \frac{x^5}{120} - \dots \right) = x^2 - \frac{x^4}{6} + \frac{x^6}{120} - \dots$$

- 6 Use the known Taylor series for $\cos x$ to estimate $\cos(0.1)$ using the first three nonzero terms.

$$\cos(0.1) \approx 1 - \frac{(0.1)^2}{2} + \frac{(0.1)^4}{24} = 1 - 0.005 + 0.0000042 \approx 0.995004.$$

The Lagrange error bound

- 7 State the Lagrange Error Bound formula for the remainder $R_n(x)$ of a Taylor polynomial of degree n .

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}, \text{ where } M \text{ is an upper bound for } |f^{(n+1)}(z)| \text{ on the interval between } a \text{ and } x.$$

- 8 Use the Lagrange Error Bound to estimate the maximum error when approximating $\sin(0.5)$ with the Taylor polynomial $P_3(x) = x - \frac{x^3}{6}$ (degree 3, centered at 0), given $|f^{(4)}(z)| \leq 1$ for all z .

$$|R_3(0.5)| \leq \frac{1}{4!} |0.5|^4 = \frac{0.0625}{24} \approx 0.0026.$$

Radius of convergence of Taylor series

- 9 State the radius of convergence of the Maclaurin series for e^x , $\sin x$, and $\cos x$.
All three series have radius of convergence $R = \infty$ — they converge for all real x .
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Visualizing Taylor polynomial approximations

- 10 Compare $f(x) = \sin x$ to its degree-5 Maclaurin polynomial $P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ on $[-\pi, \pi]$, and describe where the approximation is best and worst.

The approximation is excellent near $x = 0$ (where the curves are nearly indistinguishable) and progressively worsens toward $x = \pm \pi$, since Taylor polynomials are most accurate near the center of expansion and lose accuracy farther away.

- 11 Compare $f(x) = e^x$ to its degree-2 Maclaurin polynomial $P_2(x) = 1 + x + \frac{x^2}{2}$ on $[-2, 2]$, and identify visually where P_2 underestimates or overestimates e^x .

Since e^x is convex (concave up) and its Taylor polynomial is tangent to it at $x = 0$ up to second order, $P_2(x)$ tends to underestimate e^x away from the center, with the gap growing faster on the positive side due to the exponential's rapid growth.

Building Taylor series at a nonzero center

- 12 Find the Taylor series for $f(x) = \cos x$ centered at $a = \frac{\pi}{2}$, up through the cubic term.

$f(\frac{\pi}{2}) = 0$; $f'(x) = -\sin x$, $f'(\frac{\pi}{2}) = -1$; $f''(x) = -\cos x$, $f''(\frac{\pi}{2}) = 0$; $f'''(x) = \sin x$, $f'''(\frac{\pi}{2}) = 1$. Series:
 $-(x - \frac{\pi}{2}) + \frac{1}{6}(x - \frac{\pi}{2})^3 - \dots$

Applying the Lagrange Error Bound to bound accuracy requirements

- 13 Using the degree-4 Maclaurin polynomial for $\cos x$ and the bound $|f^{(5)}(z)| \leq 1$, find the maximum error in approximating $\cos(0.3)$.

$|R_4(0.3)| \leq \frac{1}{5!} |0.3|^5 = \frac{0.00243}{120} \approx 0.0000203$ — an extremely small error, confirming the Maclaurin polynomial is highly accurate this close to the center.

The binomial series

- 14 State the binomial series for $(1+x)^k$ up through the quadratic term, and use it to find the Maclaurin series for $\sqrt{1+x}$ (i.e. $k = \frac{1}{2}$) up through the quadratic term.

$(1+x)^k = 1 + kx + \frac{k(k-1)}{2}x^2 + \dots$. With $k = \frac{1}{2}$: $\sqrt{1+x} = 1 + \frac{1}{2}x + \frac{(1/2)(-1/2)}{2}x^2 + \dots = 1 + \frac{x}{2} - \frac{x^2}{8} + \dots$.

- 15 Use the series found above to approximate $\sqrt{1.1}$ using the first three terms.

$\sqrt{1.1} \approx 1 + \frac{0.1}{2} - \frac{(0.1)^2}{8} = 1 + 0.05 - 0.00125 = 1.04875$ (true value ≈ 1.04881).

Deriving new Maclaurin series from known ones

16 Use the known series for e^x to find the Maclaurin series for $\cosh x = \frac{e^x + e^{-x}}{2}$, up through the quartic term.

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots; e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} - \cdots. \text{ Averaging cancels odd terms:}$$

$$\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \cdots.$$